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DEMONSTRATE AN UNDERSTANDING OF PREVENTATIVE MAINTENANCE, ENVIRONMENTAL AND SAFETY ISSUES IN A COMPUTER ENVIRONMENT

US 114636

SECTION B: SUMMATIVE ASSESSMENT

QUESTION 1

Overheating causes hardware failure, so maintenance is needed to prevent it:

1. Component damage - Excess heat warps solder joints, damages CPU, GPU, RAM, and shortens lifespan.

2. Thermal throttling - The computer slows down to reduce heat, causing lag and poor performance.

3. Automatic shutdowns - The system shuts off to prevent permanent damage, leading to data loss.

4. Fan failure- Fans work harder and wear out faster, requiring replacement.

5. Dust buildup - Heat attracts more dust which blocks vents, making overheating worse.

QUESTION 2

Since Thabo used his laptop outside at parties:

1. Blocks ventilation- Dust clogs fans and vents, causing overheating and shutdowns.

2. Causes short circuits - Dust particles can conduct electricity and damage motherboards and other internal components.

3. Reduces performance- Overheating from dust leads to throttling, freezing, and eventual hardware malfunction.

QUESTION 3

I. Types of fires

- Type A: Fires involving ordinary combustibles like paper, wood, cloth, and furniture.

- Type B: Fires involving flammable liquids such as solvents, oils, and cleaning chemicals.

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- Type C: Electrical fires involving energized electrical equipment like computers, servers, cables, and power supplies.

II. CO2 or Dry Chemical extinguisher (Class C) * should be used.

- CO2 extinguishers are best for electrical fires because they don't leave residue and won't damage electronics.

- Dry Powder extinguishers can also be used for A, B, and C fires.

- Never use water on electrical fires as it conducts electricity and can cause electrocution.

QUESTION 4

•Exercising the eyes by periodically focusing on objects at varying distances.

•Blinking regularly.

•Keeping the air around you moist

•Adjusting the screen height or seating so that when sitting comfortably your eyes are in line with the top of the monitor screen.

•Adjusting the brightness control on your monitor for comfort.

•Adjusting the contrast on your monitor to make the characters distinct from the background.

•Adjusting the refresh rate of your monitor to stop it flickering.

•Positioning monitors to avoid glare.

•Keeping your monitor screen clean.

•Keeping the screen and document holder at the same distance from your eyes.

•Servicing, repairing or replacing monitors that flicker or have I adequate clarity.

•Regular eye testing.

QUESTION 5

Advise Sam to:

1. Do NOT throw in general trash- Batteries contain toxic chemicals like lithium, lead, cadmium.

2. Take to certified e-waste collection point- Most electronics stores and municipalities have battery bins.

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3. Use manufacturer take-back program- Many laptop brands recycle old batteries.

4. Tape battery terminals - Prevents short circuits and fires during transport.

5. Store in cool, dry place before disposal to avoid leakage.

6. Follow local environmental regulations for hazardous waste disposal.

QUESTION 6

1. Seal in bag - Prevent toner powder from leaking which is harmful if inhaled.

2. Return to manufacturer or recycling program - Most brands like HP, Canon offer free cartridge recycling.

3. Take to e-waste recycling center- Ensures proper handling of plastics and residual toner so it doesn't pollute soil and water.